

Pranav Karawale

Precision *beyond* pixels.

pranav@karawale.com | +91 86051 49229 | Navi Mumbai, IN
karawale.com | github.com/retronav | linkedin.com/in/pranavkarawale

SUMMARY

Founding Engineer and CS student building edge-deployed IoT pipelines and applied ML systems for industrial clients. Proven track record of scaling consumer applications and optimizing cloud infrastructure.

SKILLS

Languages: Python, TypeScript, Go, C, Bash **Infra & DevOps:** AWS (EC2, S3), Docker, Traefik, Grafana, Prometheus
Backend: Node.js, FastAPI, PostgreSQL, MongoDB, MQTT, GRPC **AI & Edge:** YOLO, OpenCV, Raspberry Pi, STM32, Modbus/PLC, PlatformIO

EXPERIENCE

Pravideon Pvt Ltd

Founding Engineer

Jan. 2024 – Present

Navi Mumbai, India

- Worked on designing AI models and controls for fire monitors to detect and respond to fire hazards autonomously. Also developed a secure dashboard for viewing real-time location feed and controlling the monitors remotely.
- Built live data pipeline using Python and PostgreSQL, indexing 15 million+ historical records in under 24 hours, and ingesting daily new data.
- Developed real-time IoT monitoring dashboard deployed across 10+ industrial sites, achieving 99.9% uptime.
- Created AWS-hosted ingestion pipeline processing and indexing 200K+ rich-media questions for enterprise LMS.

Avignon Université

Research Intern

Jun. 2025 – Aug. 2025

Avignon, France

Designed new authenticated encryption with associated data encapsulation scheme for LoRaWAN for protection against various MiTM, replay attacks, and developed vector database based reinforcement learning for auditing end devices.

Freelancing

AI Tooling Engineer

Apr. 2023 – Dec. 2023

Remote

Architected a LLM agnostic prompt orchestration framework utilizing React JSX as a declarative language to build, manage, and scale complex agentic workflows across different LLM providers (OpenAI, Anthropic, Gemini).

EDUCATION

Vishwaniketan's iMEET (University of Mumbai)

B.E. Computer Engineering

Nov. 2022 – Jun. 2026

PROJECTS

Newage Fire AI

Autonomous Fire detection and response system. Uses Modbus and custom trained models to target and extinguish fires using industrial monitors, and send live alerts.

Ixora <http://github.com/retronav/ixora>

An extension pack to make interactive WYSIWYG markdown editors using CodeMirror 6.

DuckThisFile

Self-hosted file portal with microservice architecture for image and PDF processing with privacy.

ACHIEVEMENTS

Winner: Echelon '25 Hackathon (AI P&ID Digitization)

Gold Medallist: JuniorSkills India 2021 (Web Technologies - National Level)

Leadership: Technical Lead, GDG on Campus VIMEET (2024-25)

OSS Contributor: Astro [<https://astro.build>]